Periodic Trends

Answer key with worked solutions

A. Understanding

1. Effective nuclear charge.

> The net positive charge felt by an outer electron after shielding by the inner electrons. (a) Na: 11 − 10 = **1** (b) S: 16 − 10 = **6**

2. Radius across a period.

> Across a period protons are added, but the electrons go into the same outer shell and the number of inner (shielding) electrons stays the same. Z_{eff} increases, so the outer shell is pulled closer to the nucleus.

B. Ranking

3. Atomic radius.

> In period 3, radius decreases from left to right (S < Mg < Na); Ca is below Mg with an extra shell. **S < Mg < Na < Ca**

4. First ionization energy.

> Down group 1 ionization energy decreases (K < Na < Li); Ne, a noble gas at the right of period 2, is highest. **K < Na < Li < Ne**

5. Electronegativity.

> **Na (0.93) < I (2.66) < Br (2.96) < Cl (3.16)**

C. Ionization energy

6. Group 1.

> Ionization energy **decreases** down the group, from 520.2 kJ/mol (Li) to 375.7 kJ/mol (Cs). Each element has one more shell, so the outer s electron is farther from the nucleus and more shielded by inner electrons. It is held less strongly and is easier to remove.

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7. Period 2 exceptions.

> **B < Be:** boron's outer electron is in a 2p orbital, which is higher in energy than beryllium's 2s orbital, so it is easier to remove. **O < N:** nitrogen's three 2p electrons are unpaired; oxygen's fourth 2p electron must pair with another, and the repulsion between the paired electrons makes one of them easier to remove.

D. Ionic radius

8. Ions and atoms.

> (a) Cl^- is larger: the extra electron increases repulsion between electrons, and the same 17 protons now hold 18 electrons. (b) **Mg** is larger: Mg^{2+} has lost its whole third shell ($3s^2$), and its remaining 10 electrons are held by 12 protons.

E. Challenge

9. Isoelectronic series.

> $S^{2-} > Cl^- > K^+ > Ca^{2+}$. All have 18 electrons, but the number of protons increases (16, 17, 19, 20). More protons pull the same number of electrons in more tightly, so the ion is smaller.

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